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A Cost-Effective Sequential Route Recommender System for Taxi Drivers

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Abstract. This paper develops a cost-effective sequential route recommender system to provide real-time routing recommendations for vacant taxis searching for the next passenger. We propose a prediction-and-optimization framework to recommend the searching route that maximizes the expected profit of the next successful passenger pickup based on the dynamic taxi demand-supply distribution. Specifically, this system features a deep learning-based predictor that dynamically predicts the passenger pickup probability on a road segment and a recursive searching algorithm that recommends the optimal searching route. The predictor integrates a graph convolution network (GCN) to capture the spatial distribution and a long short-term memory (LSTM) to capture the temporal dynamics of taxi demand and supply. The GCN-LSTM model can accurately predict the pickup probability on a road segment with the consideration of potential taxi oversupply. Then, the dynamic distribution of pickup probability is fed into the route optimization algorithm to recommend the optimal searching routes sequentially as route inquiries emerge in the system. The recursion tree-based route optimization algorithm can significantly reduce the computational time and provide the optimal routes within seconds for real-time implementation. Finally, extensive experiments using Beijing Taxi GPS data demonstrate the effectiveness and efficiency of the proposed recommender system.

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Keywords: recommender system • deep learning • business effective strategy • route recommendation

1. Introduction

The availability of global positioning system (GPS) trajectory data with rich information of human mobility provides an unprecedented opportunity for real-time mobile recommender systems, including location-based marketing (Hopkins and Turner 2012), public transit systems (Du et al. 2019), travel tour recommendation (Ge et al. 2014), and ride-hailing service (Feng et al. 2020). Despite many studies on human mobility pattern discovery and traffic demand prediction using GPS data, taxi companies are facing a business opportunity and a technical challenge of exploiting massive taxi GPS data to provide direct guidance for vacant taxis in search of their next passenger (Ge et al. 2010). Different from the mobility-on-demand ride-hailing systems (Alonso-Mora et al. 2017, Braverman et al. 2019, Feng et al. 2020, Zhu et al. 2021), such as DiDi, Uber, and Lyft, taxi companies in urban cities, such as Beijing Taxi and NYC Yellow Cab, provide street-hail services where taxi drivers search for the next passenger through a searching route according to their experience and habit. However, because of the lack of local information on passenger distribution and surrounding taxi activities, taxi drivers are constantly concerned about the difficulties in finding the next passenger and the low profit due to competition among vacant taxis.

Recent efforts have been made to develop intelligent mobile route recommender systems for taxi drivers who offer street-hail services, aiming to optimize the searching route for vacant taxis (Xiao et al. 2020). Such a mobile route recommender system suggests a searching strategy for a taxi driver subscribed to this service, as soon as the

taxi becomes vacant and submits a route query. Existing methods can be categorized into three groups. The first group of methods provides an energy-efficient strategy of waiting for the next passenger by suggesting a list of potential parking points with the highest pickup probability (Ge et al. 2010, Yuan et al. 2013a, Xu et al. 2016, Kong et al. 2017). The second group provides a proactive strategy in search of the next passenger by suggesting a route consisting of a series of pickup locations with the shortest searching distance or searching time (Wu et al. 2017, Garg and Ranu 2018, Qu et al. 2020). The last group of methods, represented by Qu et al. (2014), focuses on optimizing the net profit of the searching route offered to a single taxi by combining the passenger pickup probability and expected profit.

Despite the previous efforts, developing a practical cost-effective route recommender system that responds in real-time remains challenging. First, to provide route recommendations, an accurate prediction of passenger pickup probability is desired at the road segment level. Existing studies of traffic demand prediction, including the autoregressive methods (Li et al. 2012, Moreira-Matias et al. 2013), recurrent neural networks (Xu et al. 2017), and graph convolution network (GCN)-based predictors (Jiang and Luo 2021), were developed based on the temporal or spatial dependencies of traffic speed or traffic flow. However, the taxi route recommendation requires a dynamic passenger pickup predictor, which remains technically challenging because of multiple impact factors, such as time, road network, weather conditions, traffic conditions, and simultaneous modeling of taxi supply and demand. Second, the pickup probability predictor needs to be embedded into a route optimization strategy to maximize the expected profit. Because of the large route searching space in a complex road network and a large number of taxis requesting recommendation solutions, a real-time sequential route searching algorithm remains unexplored. It is also vital to avoid *recommendation overload*, which is a phenomenon due to the repeated recommendations of the same searching route to too many taxi drivers, causing taxi congestion, and profit decline (Qu et al. 2014, Xiao et al. 2020). However, existing taxi recommender systems, including the aforementioned energy-efficient, shortest-route, and cost-effective recommender systems, are supported by offline recommendation algorithms that do not carefully address the issue of recommendation overload.

To address the aforementioned technical challenges, we propose a new cost-effective sequential route recommendation method, combining a deep learning-based prediction model for road segment-level pickup probability predictions and a recursive route searching algorithm for real-time optimal route recommendations. Specifically, the deep learning-based prediction model is based on a GCN and long short-term memory (LSTM) to capture the dynamic human mobility patterns and taxi demand-supply distribution from both topological and temporal perspectives. Meanwhile, context information, such as weather conditions, is also considered to improve prediction accuracy. Different from the existing GCN-based traffic demand or speed prediction methods (Jiang and Luo 2021), the proposed GCN-LSTM model is developed for passenger pickup probability prediction, which integrates the long-term and short-term patterns of taxi demand-supply distribution in a topological road network. As such, it captures the neighborhood taxi searching behavior and customer demand changes caused by previous recommendations, enabling a more timely and precise prediction of pickup probability that dynamically balances the taxi demand and supply to avoid recommendation overload. Furthermore, to improve computational efficiency for route optimization, we integrate the pickup probability predictor with a recursive route searching algorithm to support real-time route recommendations for a sequence of taxi drivers. The pickup probabilities are updated dynamically according to surrounding taxis' previously scheduled driving routes to avoid recommendation overload. We take the distribution of pickup probability as an input to estimate the potential profit of candidate searching routes represented by a recursion tree structure, significantly reducing the size of route searching space for real-time recommendations. As such, we can serve more than 3,200 taxis in a large road network consisting of 18,301 road segments, as demonstrated in our numerical experiments using the Beijing Taxi GPS data. The performance of the proposed system sets a new benchmark against the state-of-the-art route recommender systems for taxi drivers and confirms its effectiveness in addressing the recommendation overload issue.

The remainder of this paper is organized as follows. Section 2 summarizes the related literature. In Section 3, we present the proposed sequential route recommender system framework. Sections 4 and 5 introduce in detail the proposed dynamic pickup probability prediction model and sequential route optimization model, respectively. Numerical experiments using real-world data from Beijing Taxi are reported in Section 6. Finally, Section 7 concludes this work.

2. Related Work

The recent advances in wireless sensor technologies in mobile environments have enabled extensive mobile applications, which have motivated tremendous efforts to improve work efficiency and effectiveness for taxi drivers

(Zhu et al. 2014). The related literature mainly focuses on traffic demand prediction and route recommender systems, to be reviewed separately in another article.

2.1. Traffic Demand Prediction

The prediction of traffic demand has been investigated as a key to the success of intelligent transportation systems (Yuan and Li 2021). Early studies of traffic demand were focused on understanding the spatiotemporal patterns of taxi trips and taxi driving behavior (Hochmair 2016, Xu et al. 2016). Detecting hot spots of taxi traffic has been studied for taxi relocation, aiming to find the regions with high taxi demand (Moreira-Matias et al. 2013, Xu et al. 2017, Yao et al. 2018) or extreme traffic conditions (Li et al. 2021). In terms of forecasting methodologies, one line of the literature considers traffic demand prediction as a time series forecasting problem and builds on autoregressive integrated moving average (ARIMA)-based autoregressive models to predict passenger demands in certain hot spots and taxi stands (Li et al. 2012, Moreira-Matias et al. 2013). Chang et al. (2010) improved taxi demand prediction accuracy by integrating context information of the time, weather, and taxi location. The other line of related work uses machine learning and deep sequence models to improve prediction accuracy. Xu et al. (2017) and Zhang et al. (2022) implemented recurrent neural networks for taxi demand hot spots detection. Yu et al. (2017) built a deep neural network based on LSTM for traffic prediction in extreme conditions. A deep multiview spatial-temporal network model by integrating the LSTM model and the convolutional neural network (CNN) was developed for taxi pickup demand in big regions (0.7 km × 0.7 km; Yao et al. 2018).

More recently, the success of graph neural networks (GNNs) has shown superior performance on modeling graph-based networks (Wu et al. 2020a), such as document citation networks (Thekumparampil et al. 2018, Liu et al. 2020), social networks (Fan et al. 2019, Ma et al. 2021), and network-based traffic demand prediction (Jiang and Luo 2021). With the great potential in road network modeling, the gated attention networks (GaANs) (Zhang et al. 2018) and the multivariate graph neural network (MTGNN) (Wu et al. 2020b) demonstrated the advances of graph modeling in traffic flow prediction. A newly emerging GNN-based model is the GCN, which enables the modeling of spatial distribution as topological graphs (Kipf and Welling 2017). The GCN has excellent advantages for spatial traffic demand analysis in a road network once combined with the temporal dynamic property. Lin et al. (2018) implemented GCN to learn hidden heterogeneous pairwise correlations between bike stations to predict station-level traffic demand in a bike-sharing network. Other GCN variants were proposed to study traffic flow with a multigraph convolutional neural network model (Chai et al. 2018) and an attention-based spatial-temporal graph convolutional network (Guo et al. 2019). Du et al. (2020) proposed a dynamic transition GCN to simultaneously extract temporal and spatial features to obtain accurate demand predictions for taxi transition demand between origin-destination (OD) pairs.

Indeed, existing regional traffic speed and traffic demand prediction methods have built the foundation for intelligent mobile recommendation systems that suggest the optimal routes between OD pairs with the minimum driving time or the hot spots with large taxi demand. However, because of the following three limitations, they are not directly applicable to the cost-effective route recommendation that maximizes taxi profit. First, the current autoregressive methods, recurrent neural networks, and GCN-based methods were developed based on the temporal or spatial dependencies of one single factor of either traffic speed or traffic flow. However, maximizing the expected profit of a searching route depends on an accurate pickup probability estimation, which requires simultaneous modeling of taxi supply and demand distribution. The supply distribution depicts taxis' searching behaviors and the demand distribution profiles the locations of future customers. Second, existing GCN-based methods were developed to model the short-term dependencies of traffic demand. As both taxis' searching behavior and customers' demand distribution have daily and hourly patterns, it is desirable to capture the long-term dependencies in the road network in addition to the short-term dependencies. Finally, existing mobile recommender systems have not considered how recommendations made to current taxis would change future pickup probability. To wit, Qu et al. (2014) first pointed out the importance of taxi pickup probability on road segments for the cost-effective route recommender systems; Qu et al. (2014) and Sun et al. (2017) only considered a simple historical mean method to estimate pickup probability; and Yuan et al. (2011) considered the spatial dependency of pickup probability in road clusters. To close the aforementioned research gaps and improve the accuracy of pickup probability prediction, this paper extends the GCN-based methods by integrating the long-term and short-term patterns of taxi demand-supply distributions in a topological road network, as well as by considering the mutual dependence of current recommendations and future pickup probability distributions.

2.2. Mobile Route Recommender System

The route recommender systems for taxi drivers aim to leverage historical taxi GPS traces for improving recommendation performance. It has been formulated with various objectives to provide routing suggestions for a vacant taxi searching for the next passenger. Ge et al. (2010) developed an energy-efficient mobile recommender system

for an optimal sequence of waiting points for the next passenger. Yuan et al. (2010, 2011, 2013a, b) and Ge et al. (2021) worked on a series of intelligent taxi services, such as pickup point detection and driving direction recommendation. Kong et al. (2017) further developed a recommender system with a time-location combined service recommendation to provide top N recommended passenger searching areas. However, these studies only focus on pickup point recommendations while ignoring the potential profit of exact searching routes. Qu et al. (2014) first provided an entire driving route instead of discrete pickup points based on the expected profit for taxi drivers. Similar research has been conducted to recommend searching routes for taxi drivers with the objective of minimizing the waiting time or the searching distance (Wu et al. 2017, Garg and Ranu 2018, Qu et al. 2020, Ye et al. 2019). Ye et al. (2018) compared the recommender systems with different objective functions, and Wang et al. (2019) found that taxi drivers could improve hourly earnings by selecting long trips rather than aiming for cruising-time reduction. However, both the relocation and route recommendation did not consider the recommendation overload issue in a real-time environment for a sequence of vacant taxi drivers searching in the same region.

2.3. Recommendation Overload

Traditional recommender systems may repeatedly recommend the same or similar items to a group of customers, resulting in system-generated best-selling products and a low recommendation diversity that decreases users' purchase intention and recommendation effectiveness. Similarly, recommendation overload can happen in a route recommender system if the same route is recommended to too many drivers in the same area (Qu et al. 2014). Qu et al. (2014) introduced a top K sequential recommendation framework that compares trajectory similarities between subsequently recommended routes to mitigate the influence of prior recommendations made to surrounding taxis. The top K strategy avoids recommending the same route repeatedly, but it does not consider the profit difference on alternative routes and may direct taxis to a route with a low expected profit. Xiao et al. (2020) proposed a multi-user mobile sequential recommendation system that recommends nonoverlapped searching routes to a group of taxis with a road capacity constraint. However, this system assumes that taxis start searching simultaneously and is not applicable for sequential recommendations when taxis come in a sequence. In real-world practices, a recommendation should be made within seconds of a submitted query. Thus, the demand and supply distribution can change as recommended routes are sent to taxi drivers. However, the existing route recommender systems do not incorporate recommended searching routes for future route recommendations. In this paper, we will achieve a balanced taxi supply and demand distribution by considering the potential recommendation overload.

3. Sequential Route Recommender System Framework

This section introduces the preliminaries and research data and then presents the key research modules of the sequential route recommender system.

3.1. Preliminaries

We provide some basic concepts and definitions of terminologies, which will be used throughout this paper. The sets and parameters of the problem under study are summarized in Table 1 and will be explained in more detail subsequently. Additional notation specific to the deep learning model and the route searching algorithm will be introduced in their respective sections later.

Definition 1 (Road Segment). A road segment r associated with two junctions (such as an intersection, a roundabout, and a dead end) is the unit component of a driving route. It has several static properties including the one-way driving direction and segment length. For each period t, a road segment r can be used as the basic sensor for recording the number of pickup orders $o_{r,t}$ and the number of vacant taxis $v_{r,t}$ as the spatial distribution of taxi demand and supply, respectively. Each segment r has several adjacent segments directly connected to it. We define road segment r's adjacent segments $\mathcal{A}_r$ as the segments to which a taxi can drive directly from r. The travel time t_r for the road segment r is estimated by dividing the segment length by the average driving speed of taxis on segment r within the same time period from the historical data. If the historical data are missing for a segment, a 20-km/h driving speed is used to calculate the travel time.

Definition 2 (Road Network). A road network $\mathcal{G}_{r_0} =< \mathcal{V}, \mathcal{E} >$ is a directed graph with the searching-start road segment r_0 at the center, where $\mathcal{V}$ represents the node set consisting of all road segments in this network, and the edge (r_i, r_j) in set $\mathcal{E}$ records the direct connection from road segment r_i to r_j. The road network $\mathcal{G}_{r_0}$ consists of all road segments that the vacant taxi can visit from the searching-start segment r_0.

Definition 3 (Searching Route). A searching route R_{r_0} is represented by a sequence of road segments $R_{r_0} =< r_0, r_1, \ldots, r_n >$, where n is the number of road segments on this route following r_0. Starting from road segment r_0, a taxi

Table 1. Notation

Sets	Description
$\mathcal{A}_r$	Set of road segments adjacent to road segment r
R_r	A searching route starting from road segment r
$\mathcal{G}_r$	A road network with the searching-start road segment r at the center
$\mathcal{S}_r$	Set of searching routes starting from road segment r
$\mathcal{K}$	A queue of taxis requesting route recommendations
Υ_r	Recursion tree with searching-start road segment r as the root node
$G^s(R;r,t)$	Expected profit by searching along route R starting from r at time t
$G^w(R;r,t)$	Expected profit by driving to the virtual station along route R and waiting for the next passenger
Parameters	
n	Number of road segments in a route
N	Number of road segments in the road network
d	Number of input features for pickup probability prediction
p_r	Pickup probability on road segment r
e_r	Expected earning per pickup event on road segment r
t_r	Driving time along road segment r
T_R	Driving time along route R
T_r	Waiting time at the virtual station on road segment r
$v_{r,t}$	Number of vacant taxis on road segment r at time period t
$o_{r,t}$	Number of pickup events on road segment r at time period t
T	Searching time limit

driver can choose several searching routes, which form a route candidate set $\mathcal{S}_{r_0}$. Each eligible route in $\mathcal{S}_{r_0}$ is bounded by a maximum searching time τ (e.g., 10 minutes as used in this paper), that is, $t_{r_0} + t_{r_1} + \cdots + t_{r_n} \le \tau$. Hence, n can be different for different routes in $\mathcal{S}_{r_0}$.

Definition 4 (Sequential Route Recommendation). A mobile route recommendation system aims to recommend a searching route for a vacant taxi in search of the next passenger. Furthermore, the route recommendations are made sequentially as route enquiries emerge in the system. Although optimizing the search routes over a group of taxis can be superior to optimizing over the same group sequentially, real-world practices require a recommendation to be made within seconds whenever a route query is received. Thus, waiting for a buffer of route queries may result in a high query response latency, except for the very rare case when a group of queries emerges at the same time in the same region.

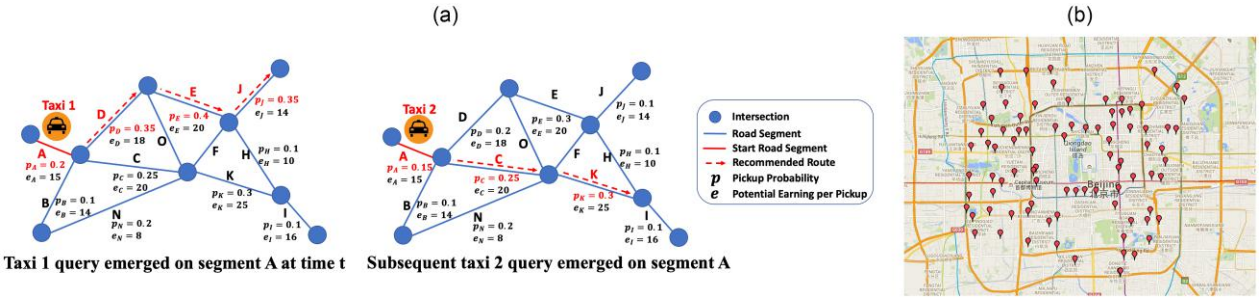
As taxi drivers highly prefer searching routes with high pickup probabilities and high earnings per trip (Wang et al. 2019), we consider the next pickup event on a road segment $r \in R_{r_0}$ with a probability p_r and an expected earning e_r from this pickup. Here, e_r is estimated by averaging earnings in historical pickup events that happened on road segment r (Qu et al. 2014). Then, the expected profit (earning), $G^s(R;r_0,t_0)$, of searching along a given route $R =< r_0, r_1, \ldots, r_n >$, starting at road segment r_0 and time t_0, is defined as follows:

$$G^s(R;r_0,t_0) = p_{r_0}e_{r_0} + \sum_{i=1}^{n}\prod_{j=0}^{i-1}(1-p_{r_j})p_{r_i}e_{r_i}, \tag{1}$$

where the expected profit of picking up a passenger on road segment r_i ($r_i \in R$) is defined as $\prod_{j=0}^{i-1}(1-p_{r_j})p_{r_i}e_{r_i}$, considering the probability that a taxi does not find a passenger until reaching road segment r_i. Hence, the sequential route recommendation problem searches for an optimal route $R \in \mathcal{S}_{r_0}$ that maximizes $G^s(R;r_0,t_0)$. Because all eligible routes in $\mathcal{S}_{r_0}$ are bounded by the maximum searching time τ, the mobile route recommendation problem under study aims to provide a searching route with the maximum expected profit for a taxi within a searching time limit of τ. A larger value of τ generally allows a longer searching path with more road segments n and thus a higher expected profit. However, because the expected earning on a road segment is discounted by the probability term in Equation (1), the marginal value of the added expected profit for increasing τ will diminish. Qu et al. (2014) suggested a fixed number of searching road segments $n = 5$. To make a fair comparison of expected earnings per minute, we consider a fixed searching time limit of 10 minutes in our numerical experiment in Section 6.

To illustrate the above definition of the sequential route recommendation problem, Figure 1(a) presents a scenario where two taxis arrive at road segment A in sequence and are recommended a searching route each. In the left subfigure of Figure 1(a), taxi 1 arrives at road segment A at time t and is recommended an optimal route "A-D-E-J" based on the estimated pickup probabilities on all road segments. Consequently, taxi 1 becomes an active supply along this route, which decreases the pickup probabilities on road segments A, D, E and J. In the right

Figure 1. (Color online) Example of Different Route Recommendation Strategies



Notes. (a) Searching route recommendation. (b) Virtual station candidates.

subfigure, when taxi 2 submits a query while taxi 1 is still searching along route "A-D-E-J," this route may become suboptimal due to the decreases in pickup probabilities. Thus, another route "A-C-K" becomes optimal and is recommended to taxi 2.

As seen from the previous example, an intelligent route recommendation system should provide alternative searching routes for a sequence of vacant taxis competing in the same region, whenever one particular route has been overly recommended. In this paper, we incorporate the searching routes of the previously recommended taxis to update the pickup probability distribution, so as to help the subsequent taxis avoid recommendation overload.

Definition 5 (Virtual Station). Taxi, as a station-free transportation mode, can have arbitrary passenger pickup locations scattered around urban areas. During certain periods, traffic demands tend to congregate in certain areas, such as a school entrance when the school is over or a residential area during rush hours. Experienced taxi drivers can identify the virtual stations where they can wait for the next passenger rather than driving along a route to search for a passenger (Ge et al. 2010). Accordingly, we identify a set of virtual stations on certain road segments where taxi drivers may stop and wait for the next passenger. A virtual station is active when it is more profitable waiting than actively searching along a route, and thus may only be active during certain time periods. For example, a residential entrance is an active virtual station during morning rush hours while staying inactive during other time periods.

When adopting the waiting strategy, taxi drivers choose to give up the potential profit of searching along a route. Therefore, given the current location r_0 of a taxi driver and the location of a selected virtual station r, the potential net profit of the waiting strategy $G^w(R; r_0, t_0)$ can be computed as follows:

$$G^w(R; r_0, t_0) = \frac{1}{T_R + T_r} e_r,$$

where R represents the shortest route from r_0 to the virtual station r, T_R is the attendant driving time, and T_r is the extra time the taxi driver spends in a waiting line at virtual station r for the next passenger. Figure 1(b) presents the distribution of 90 virtual station candidates in our data set, comprised of taxi stands and pickup hot spots. These virtual stations are extracted with the following criteria: (1) the pickup frequency is larger than 100 at this location, and (2) taxis wait for more than three minutes with a driving speed of zero followed by a pickup event during peak hours.

3.2. Research Data

We study the route recommendation problem for Beijing Taxi. Summary statistics of the Taxi GPS trace data, the Beijing City road network data, and the meteorology data used in this study are presented in Table 2. The research data can be found at the GitHub site of this paper (Liu et al. 2023).

3.2.1. Road Network. The road network of the urban areas in Beijing City consists of 18,301 road segments, indexed by two vertices on a single segment. Two or more segments are connected if they share the same vertex. A road segment is one-way if the driving directions of all taxis passing through this segment are identical.

3.2.2. Taxi GPS Traces. Each GPS record has the information of taxi ID, GPS location, service status (occupied or vacant), fare identifier for midnight and airport surcharge, driving direction, driving speed, and time stamp. The GPS trace is recorded at a maximum frequency of every three seconds. The research data are derived from the taxi

Table 2. Summary Statistics of the Data Sets

Data source	Data information	24-hour full set	8 a.m. to 11 a.m.	1 p.m. to 3 p.m.	5 p.m. to 8 p.m.
Taxi GPS traces	No. of road segments	18,301	18,301	18,301	18,301
	No. of in-service taxis	3,258	3,250	3,249	3,256
	No. of GPS points	88.98 million	11.15 million	7.05 million	11.08 million
	No. of pickup events	1.35 million	0.18 million	0.12 million	0.23 million
	Profit per trip	24.17 (±17.03)	25.28 (±18.94)	23.07 (±17.55)	23.57 (±17.21)
Meteorology data	Temperature	[26.08, 57.79] °F	[26.08, 51.22] °F	[35.56, 57.79] °F	[35.56, 54.51] °F
	Visibility	[0.99, 6.21] mile	[2.80, 6.21] mile	[1.24, 6.21] mile	[0.99, 6.21] mile
	Wind speed	[0.0, 13.42] mph	[1.04, 8.96] mph	[1.04, 6.34] mph	[0.00, 4.69] mph
	Humidity	[15%, 79%]	[19%, 70%]	[15%, 58%]	[17%, 78%]

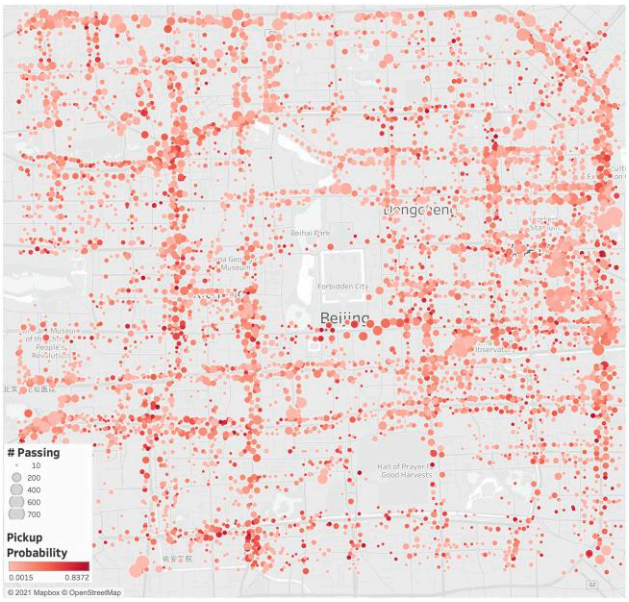
GPS traces. We identify the first "occupied" GPS point after a series of "vacant" GPS points as a record of pickup event, and the first vacant GPS point after a series of occupied GPS points as a record of drop-off event. The pickup probability of a road segment during period t is then defined as the number of pickup events divided by the total number of vacant passing-by taxis. For each time period t in the historical data, on a road segment r, we count the number of pickup orders $o_{r,t}$ up to time t and the number of vacant taxis $v_{r,t}$ to represent the taxi demand and supply on road segment r at time t. Thus, the values of $o_{r,t}$ and $v_{r,t}$ over the entire road network represent the spatial distribution of taxi demand and supply, respectively, at time t. If a vacant taxi drives through the same road segment multiple times within a single period, we count it as one single supply. Based on this historical data, we can train prediction models to forecast the pickup probability on each road segment at each time period, given the available taxi supplies on each road. More than 3,200 in-service taxis are included with their GPS traces recorded during a seven-day period. In total, our data set has 88.98 million GPS points and 1.35 million pickup and drop-off activities.

We present the taxi activities in our data set during morning rush hours in Figure 2. Each dot in Figure 2 represents a road segment with its size representing the number of passing by vacant taxis and the color representing the pickup probabilities. It is seen that the pickup probability is low for some road segments with large taxi supplies. Most taxi drivers suffer from a low pickup probability or short trips. Even for the taxi drivers who prefer searching or waiting in the highly demanded regions, their pickup probability can be low due to taxi oversupply, highlighting the importance of considering recommendation overload in this paper.

3.3. System Framework

The overarching goal of the cost-effective sequential route recommendation (C-SRR) problem is to optimize the driving routes for a sequence of vacant taxis. Once a route query from a taxi driver is received at time t_0 from a

Figure 2. (Color online) Distribution of Taxi Activities During Morning Rush Hours (8 a.m. to 11 a.m.)



searching-start segment r_0, the optimal route needs to be recommended immediately without the knowledge of any subsequent queries from other drivers. In this paper, we propose a new combined forecasting and optimization framework consisting of a dynamic pickup probability predictor and a cost-effective route recommendation algorithm.

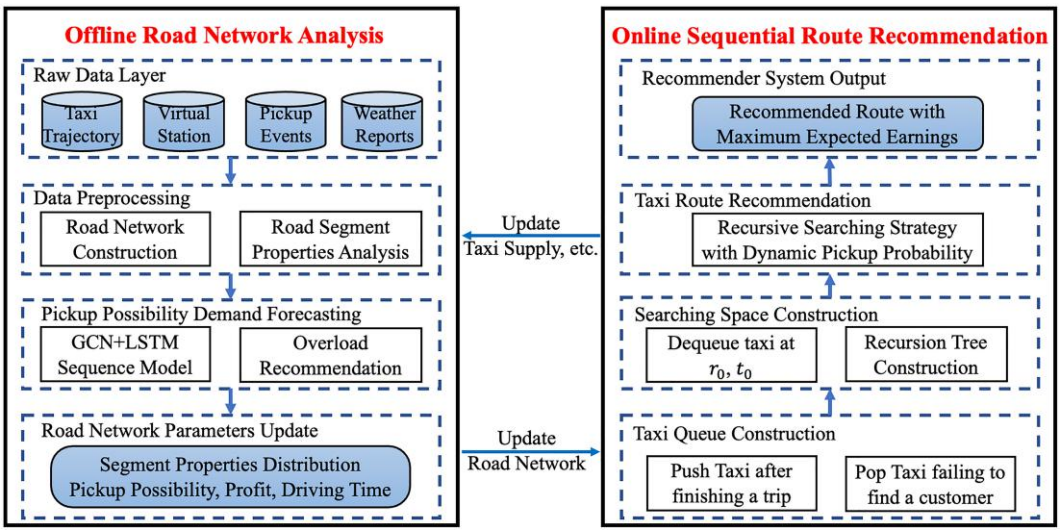
- **Dynamic Pickup Probability Prediction.** An accurate prediction of road segment-level pickup probability is essential for a balanced route recommendation. The main challenges are to fully utilize the time dynamics of human mobility, and to embed the taxi supply distribution in a road network for an accurate prediction.
- **Cost-Effective Route Recommendation.** Given a queue of taxis $\mathcal{K}$ requesting route recommendation, their query time $\{t_k|k \in \mathcal{K}\}$, query road segment $\{r_k|k \in \mathcal{K}\}$, and the pickup probability distributions updated at the query time, we aim to recommend the optimal searching route for each query (taxi) with the goal of maximizing the expected profit for the next successful pickup event within time period τ. Formally, for the kth query starting at t_k on road segment r_k, we aim to find the optimal cost-effective route $R^*_{r_k}$ from the route candidate set $\mathcal{S}_{r_k}$ by maximizing the expected profit:

$$R^*_{r_k} = \underset{R_{r_k} \in \mathcal{S}_{r_k}, m \in \{w,s\}}{\arg\max} \; G^m(R_{r_k}; r_k, t_k). \tag{2}$$

These two research modules are interconnected. The pickup probability and its associated expected profit are essential inputs for the route recommendation. Meanwhile, the route recommended to a driver can affect the taxi supply along the route and its corresponding pickup probability. In addition, there is a possibility that a taxi fails to find any passengers after searching along a recommended route. To address these practical challenges, we propose a cost-effective sequential recommender system considering recommendation overload and recommendation restart. Figure 3 presents the framework of the proposed mobile route recommender system, which provides the optimal searching routes for a queue of vacant taxis according to their searching-start time or when they complete the previous searching route without finding a passenger.

There are four steps to update the properties of road segments within a road network. First, the up-to-date historical raw data are used to provide the ground-truth for pickup probability demand predictor training, testing, and evaluation. Once the pickup probability predictor is trained, it will be used to update the future properties of road segments for online learning to avoid recommendation overload. The online module maintains a queue of taxi drivers searching for the next passenger. In general, the route queries are processed in a first in, first out (FIFO) order. A taxi that starts searching after a drop-off will be pushed to the end of the queue, whereas a taxi that fails to find a passenger after finishing the previous search route will be popped to the head of the queue. The recommender system keeps processing the first request in the queue and constructs its searching network according to its current time t_0 and road segment r_0. Then, the single taxi route recommendation algorithm is implemented to provide the optimal route. A parameter update request will be sent to the offline analyzer to update the taxi supply along the recommended searching route to improve the online recommendation performance by avoiding recommendation overload.

Figure 3. (Color online) Framework of Sequential Recommender System for Taxi Drivers



4. Dynamic Passenger Pickup Probability Prediction

The pickup probability of each road segment captures how likely a taxi driving through the road segment would find a passenger. It can be affected by the spatial-temporal dynamics of taxi activities and human mobility. Specifically, in metropolitan areas such as Beijing, it is frequently observed that demands for taxi services concentrate on hot-spot areas and decrease progressively moving away from these spots. Meanwhile, passengers' daily mobility clearly shows different patterns, varying from peak to off-peak hours. A precise prediction of pickup probability requires the model to learn taxi activities and human mobility from dynamic graph-structured road network data, which is a challenging task.

With the recent development of the GCN, some static graph-based networks, such as document citation networks, social networks, and knowledge graphs, have been modeled with promising results (Kipf and Welling 2017). GCN learns hidden layer representations that encode both the local graph structure and the features of neighborhood nodes and has been proven to scale linearly in the number of graph edges. To develop a dynamic sequential mobile route recommender system, we propose a GCN to model the spatial distribution of taxi demand-supply in a road network as a topological graph. The GCN model, with its topology modeling features, has great potential in modeling the spatial structure required to predict the taxi pickup probability. A prominent one is its ability to model the diffusion of the node states to the neighborhood, which can be considered an abstraction of the demand distribution for taxi services in the road network. On the other hand, a recurrent neural network architecture called LSTM (Hochreiter and Schmidhuber 1997) is suitable for modeling the temporal pattern of the pickup probability. LSTM has demonstrated its power in many sequential pattern learning and time series forecasting tasks, such as speech recognition, language translation, and traffic forecasting. LSTM can capture the long-term dependencies between time-series observations and the short-term ones, which resemble the daily and hourly demand patterns for taxi services. Therefore, we propose an integrated framework based on GCN and LSTM, in which the GCN captures the spatial dependency of the taxi demand-supply distribution in a road network, and the LSTM captures the temporal dynamics in pickup probability time series.

Figure 4 provides an example of the graph representation of a road network at time t. The GCN for road segment pickup probability prediction is constructed based on the taxi demand-supply distribution in the road network at time t. On this graph, each road segment is represented by a node featured by the number of vacant taxis $v_{r,t}$ and passenger pickups $o_{r,t}$. Each directed arc represents the connection between a pair of road segments. More specifically, the GCN at time t takes the input of $X_t \in \mathbb{R}^{N\times d}$ and generates an output of G_t^L, where N is the number of road segments in the network, d is the number of features considered, and L is the total number of layers for graph convolution. Thus, X_t consists of d road segment-related features at time t, including the current time period, the number of pickup events, and the number of taxis seeking customers on a road segment within period $[t-1, t)$. By considering the pickup events and vacant taxis as inputs, the GCN predictor is designed to detect the recommendation overload when a number of taxis rush into a small region, causing a significant decrease of pickup probability. Mathematically, let the adjacency matrix of the road network be A, where A_{ij} is a binary variable that equals one if road segments I and j are connected and zero otherwise. The snapshot of the graph convolution on lth layer at time t is formulated as

$$\tilde{A} = A + I_N, \tag{3}$$

$$G_t^l = \sigma\left(\tilde{D}^{-\frac{1}{2}}\tilde{A}\tilde{D}^{-\frac{1}{2}}G_t^{l-1}\Theta\right), \tag{4}$$

Figure 4. (Color online) Example of GCN Representation of Road Network Activities

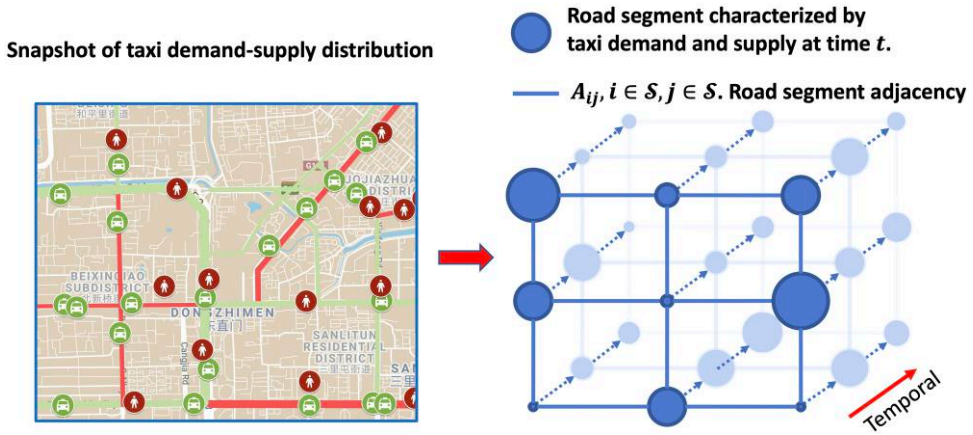
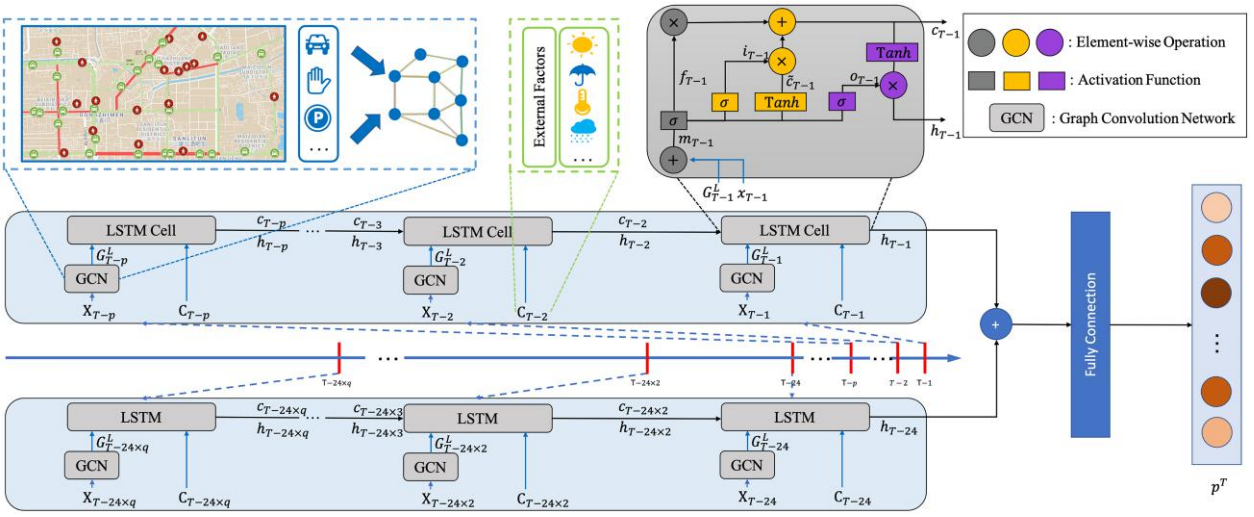


Figure 5. (Color online) GCN-LSTM Model for Road Segment Pickup Probability Forecasting



where $I_N \in \mathbb{R}^{N\times N}$ is the identity matrix and $\tilde{D}$ is the diagonal node degree matrix with each diagonal element $\tilde{D}_{ii}$ counting the number of edges for the corresponding node i. The layer-wise propagation rule in Equation (4) computes a hidden layer output G_t^l from the last layer output G_t^{l-1} and enables the model to learn the feature representations based on nodes connectivity where the information is propagated along with the neighboring nodes within the graph. The input layer $l = 0$ is the road segment-related features ($G_t^0 = X_t \in \mathbb{R}^{N\times d}$). Θ is the matrix of parameters to train and $\sigma(\cdot)$ is the rectified linear activation function which introduces the nonlinearity and improves the training efficiency. From Equation (4), the product of $G_t^{l-1}\Theta$ introduces a linear transformation for road segment features, whereas the further product with normalized adjacent matrix $\tilde{D}^{-\frac{1}{2}}\tilde{A}\tilde{D}^{-\frac{1}{2}}$ aggregates the transformed road segment features with its neighborhood. As a result, the output of GCN encodes the information of a road segment with a local neighborhood, including the intensity of taxi activities, the passengers' demand for taxi services, and the pickup probability in the road network during each time period.

Next, to capture the temporal dynamics of the road segment features into the pickup probability prediction, we integrate GCNs of different time periods into an LSTM sequence. Figure 5 shows the integrated GCN-LSTM model to predict the pickup probability at time T. As seen, the LSTM network models the temporal factors on taxi pickup probability by constructing two time series, hourly series and daily series, which, respectively, capture the short-term and long-term temporal effects. More specifically, to predict the pickup probability at time T, the LSTM takes the hourly series inputs $< m_{T-1}, m_{T-2}, \ldots, m_{T-1} >$ and the daily series inputs $< m_{T-24}, m_{T-24\times 2}, \ldots, m_{T-24\times q} >$, where p and q are the time windows for the two series, respectively, and each input m_t is the concatenation of the context information C_t and the output of GCN layer G_t^L. Here, C_t is the weather conditions at period t including the temperature, precipitation, wind, humidity, dew point, pressure, UV index, and visibility.

Mathematically, the LSTM network can be expressed as follows:

$$m_t = (C_t \oplus \text{vectorize}(G_t^L)), \tag{5}$$

$$f_t^l = \delta(W_f^l m_t + U_f^l h_{t-1}^l + b_f^l), \tag{6}$$

$$i_t^l = \delta(W_i^l m_t + U_i^l h_{t-1}^l + b_i^l), \tag{7}$$

$$o_t^l = \delta(W_o^l m_t + U_o^l h_{t-1}^l + b_o^l), \tag{8}$$

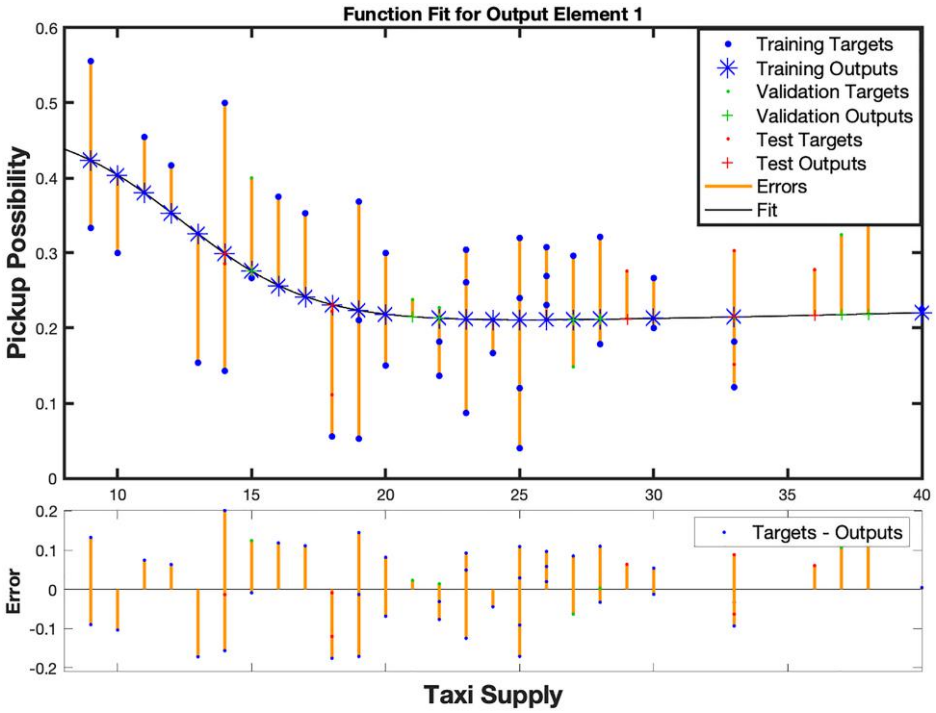
$$\tilde{c}_t^l = \tanh(W_c^l \mu_t + U_c^l h_{t-1}^l + b_c^l), \tag{9}$$

$$c_t^l = f_t^l \circ c_{t-1}^l + i_t^l \circ \tilde{c}_t^l, \tag{10}$$

$$h_t^l = o_t^l \circ \tanh(c_t^l), \tag{11}$$

where h_{t-1}^l is the hidden state of last time step, $\delta(\cdot)$ is the sigmoid function, and W, U, and b are parameters to train. Each LSTM cell has three gates: forget gate f_t^l, input gate i_t^l, and output gate o_t^l. The forget gate f_t^l controls how much information of cell state of last time step c_{t-1}^l will be added into current cell state c_t^l; the input gate i_t^l controls how much information of current cell input activation $\tilde{c}_t^l$ will be added into current cell state c_t^l; and output gate o_t^l

Figure 6. (Color online) Simulation-Based Sensitivity Analysis



controls how much information of current cell state c_t^l is used for calculating current hidden state h_t^l. Finally, the concatenation of $h_{T-1} \oplus h_{T-24}$ is fed into the fully connected layer as regression to predict the pickup probability $p^T \in \mathbb{R}^N$ at time T.

In summary, the GCN-LSTM predictor builds a nonlinear relationship between the inputs and the output, in the form of $p^T = \text{GCN-LSTM}(f_1, f_2, \ldots, f_D)$, where $f_1, f_2, \ldots, f_D$ are the taxi demand-supply distribution related features and meteorology factors. Each training sample contains the known historical pickup probability. The taxi supply information has two parts: the observed historical vacant taxi traces up to the recommendation time t_0, and the scheduled vacant taxi searching route. We construct these two kinds of taxi supply information into our demand-supply feature set to avoid recommendation overload. As a result, the predictor can proactively detect taxi oversupply by predicting a decreased pickup probability and guide the taxis to an undersupplied region via recommended routing. To illustrate this point, we simulate a scenario of taxi oversupply using the GCN-LSTM by increasing the number of vacant taxis from 8 to 40 on one road segment while keeping other parameters fixed at their average values. Figure 6 presents the fitting result of the pickup probability as a function of taxi supply, as well as its residual plot. It is seen that the pickup probability drops significantly as the taxi supply increases from 8 to 20. For a sequence of taxis, this road segment may remain in the recommended searching route until the pickup probability drops to a certain level such that future taxis only expect a small expected profit due to a reduced pickup probability. As a result, once the predictor can capture taxi oversupply from the historical data, we can leverage the predictor to avoid recommendation overload.

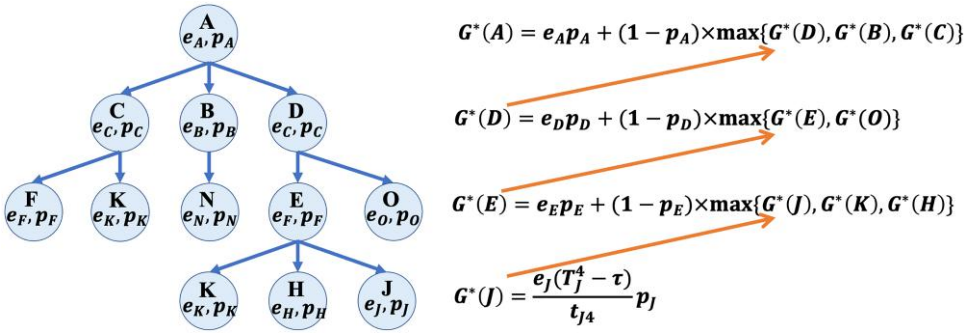
5. Sequential Route Recommendation

With the GCN-LSTM pickup probability predictor, the sequential route recommender system can be developed to sequentially optimize the searching route for vacant taxis in search of their next passenger. The sequential recommendation follows the FIFO rule, and the searching route is recommended to the taxi driver as soon as the query emerges. In this section, we first propose a recursive route searching strategy for a single taxi and then present the sequential route recommendation for a sequence of vacant taxis.

5.1. Recursive Route Searching Algorithm

Upon receiving a taxi's route query submitted on road segment r_0 at time t_0, the system collects the observed pickup events, observed passing-by events, and scheduled searching routes of nearby taxis, and then determines the best searching strategy (either route searching s or virtual station waiting w) and the corresponding driving route. The

Figure 7. (Color online) Recursion Tree Representation of a Route Network



system response time should be within seconds to provide a real-time recommendation to the driver. However, the number of route candidates increases exponentially as the number of road segments considered increases. To address the computational challenge, we propose a recursive route searching algorithm based on a recursion tree structure that transforms the network searching problem into a recursive searching problem.

Definition 6 (Recursion Tree). A recursion tree Υ_{r_0} is constructed based on the road network, where the searching-start road segment r_0 is the root node. Each node has a set of child nodes representing all connected road segments from this parent node. From node r_i in the recursion tree Υ_{r_0}, another recursion tree Υ_{r_i} is constructed starting from its root node r_i. A branch of the recursion tree will stop growing once the end node reaches the maximum searching time τ.

Figure 7 shows an example of a recursion tree Υ_{r_0} representing the route searching network shown in Figure 1(a). The searching-start road segment A is the root of the constructed recursion tree. Road segments B, C, and D are directly connected to road segment A and are represented as the child nodes of node A. After searching node A in the first layer, the searching time $T_A^1 = t_A$ does not reach the time limit τ, so we continue to build the recursion tree $\Upsilon_B, \Upsilon_C, \Upsilon_D$ from the child nodes B, C, and D. The recursion tree will stop growing once the searching time reaches the time limit. For example, the total searching time along the route "A-D-E-J" is $T_J^4 = T_E^3 + t_J$. If T_J^4 exceeds τ, the branch will then stop growing. The pickup probability at the end node (node J) is discounted by the searching time to represent a partial visit to this road segment (see $G^*(J)$ in Figure 7). The recursion tree enables a recursive searching strategy. That is, the problem of searching for the optimal route from the recursion tree Υ_A can be converted to the problem of finding the optimal routes from $\Upsilon_B, \Upsilon_C, \Upsilon_D$.

Given the estimated pickup probabilities and earnings on road segments, the recursive tree search returns the optimal route that maximizes the expected profit along the route as specified by Equation (1). In fact, the recursive tree search can be formulated under the dynamic programming framework. To this end, denote $G^*(r)$ as the maximum expected profit for a (partial) route starting at any node (road segment) r until the end of any valid searching route. From Definition 1, $\mathcal{A}_r$ is the adjacent segments connected to segment r, where $\mathcal{A}_r = \emptyset$ if the searching time reaches τ at segment r. Then, we have

$$G^*(r) = p_r e_r + (1 - p_r) \max_{r' \in \mathcal{A}_r} G^*(r'), \tag{12}$$

where the first term on the right-hand side is the expected profit on road r, and the second term is the best possible profit given a pickup does not happen on road r. It is seen that Equation (12) is the well-known Bellman's equation. Thus, the recursive tree search conforms to the principle of optimality, and returns the optimal searching route. Correspondingly, the recursive route searching algorithm (RRS) for a single taxi is presented in Algorithm 1. The RRS algorithm searches the optimal route from a constructed recursion tree Υ_{r_0}. It takes the road segment pickup possibility distribution and potential earning on the segment as an input and returns the optimal route R^* associated with a maximum expected profit $G^s(R^*; r_0, t_0) = G^*(r_0)$. The RRS algorithm stops when it reaches the end node level and returns the end node with the maximum profit. Step 5 in Algorithm 1 recursively implements the RRS algorithm starting from the next node until the stopping criteria are satisfied.

Algorithm 1 (RRS(r_0, Υ_{r_0}))
Input: a constructed recursion tree Υ_{r_0} with depth n and starting segment r_0;
Output: optimal searching route $R^*(r_0)$ and expected profit $G^*(r_0)$;
Initialize: pickup probability and earnings $\{p_r, e_r | r \in \Upsilon_{r_0}\}$
1: **if** $n = 1$ **do**

2: $R^*(r_0) =< r_0 >, G^*(r_0) = p_{r_0} e_{r_0}$
3: **else**
4: **for each** $r_i \in \mathcal{A}_{r_0}$ **do**
5: $R^*(r_i), G^*(r_i) =$**RRS**(r_i, Υ_{r_i});
6: $r_{i^*} = \arg\max_{r_i \in \mathcal{A}_{r_0}} G^*(r_i)$
7: $R^*(r_0) =< r_0, R^*(r_{i^*}) >, G^*(r_0) = p_{r_0} e_{r_0} + (1 - p_{r_0}) G^*(r_{i^*})$
8: **return** $R^*(r_0)$, $G^*(r_0)$

Figure 7 provides an example of how the RRS algorithm works. Suppose we are on road segment A as shown in Figure 1(a). To find the optimal route from A with the maximum expected profit $G^*(A)$, according to Equation (12), the solution depends on the optimal routes starting from A's neighbors B, C, and D, which recursively depend on the optimal route starting from the neighbors of B, C, and D. The process continues until reaching the end nodes, where the searching time limit is met.

Given a fully grown recursion tree Υ of depth n, where for any $r \in \Upsilon$ and $|\mathcal{A}_r| \le N$, the complexity of RSS algorithm is $\mathcal{O}(N^{n-1})$. Assuming that the computational cost of searching a recursion tree of depth n is $T(n)$, obviously, we have $T(n) \le N \times T(n-1) + N$. The computational cost of comparing the end nodes is $T(1) \le N$. Meanwhile, after computing recursively $n-1$ times, we have $T(n) \le N^{M-1} \times T(1) + \frac{N^{M-1}}{N-1}$. Therefore, the computational complexity of searching the optimal route is $\mathcal{O}(N^{n-1})$.

Considering the search for optimal routes starting from multiple roads simultaneously, the algorithm may become more effective. In the example shown in Figure 7, after the optimal route from A with the maximum profit $G^*(A)$ is found, the intermediate optimal routes starting from A's neighbors, such as $G^*(B), G^*(C), G^*(D)$, have been identified and calculated. These intermediate results can further be exploited for searching optimal routes starting from B, C, and D's neighbors, which reduces the time cost. In general, suppose we have a road network, and on each road, there is a taxi waiting for the recommended route. Here, we estimate the computational cost of the recursive searching algorithm in terms of the length n of a recursive tree. Initially, we have the expected profit $G(r;0)$ for each road, where zero represents the length of the route (the number of roads in the route except the one where the taxi currently locates). Next, Equation (12) can be exploited to find optimal routes with length 1 that gives $G(r;1)$, the time cost for this step is $\mathcal{O}(MN)$, where $M := \max_r |\mathcal{A}_r| (<< N)$ and N is the number of roads in the network. Furthermore, based on $G(r;1)$ given in the last step and Equation (12), the optimal routes with length 2 from all roads with expected profit $G(r;2)$ can be found with time cost $\mathcal{O}(MN)$, the process continues until the optimal route of length n is found. The computational complexity for multiple requests is $\mathcal{O}(nMN)$. As the computational complexity grows with the increase of the number of road segments n, a larger searching time limit τ indicates a higher computational cost due to a deeper recursion tree.

5.2. Sequential Route Recommendation

The recursive route searching algorithm can provide an optimal searching route for a single taxi. However, in real-world practices, taxis may arrive in a sequence, and the road condition, such as taxi supply, will change dynamically. For a newly emerged query, the route recommender system should consider previously scheduled taxis currently searching around to avoid recommendation overload. Hence, we provide a sequential RRS (SRRS) by integrating the dynamic pickup probability predictor and the recursive route searching algorithm for a sequence of vacant taxis.

Algorithm 2 shows the SRRS workflow. Upon receiving a query from a vacant taxi, we take a snapshot of the road network conditions and update the road network conditions, including road segment driving speed, recommended driving routes of previous vacant taxis, taxi passing-by and pickup events within the previous 60 minutes. Then, for the sequence of vacant taxis, in Step 3, we pop the head taxi i with its searching-start time t_i and current location on road segment r_i. In Steps 4 and 5, the pickup probabilities of the road segments within the recursion tree Υ_{r_i} are recalculated, from which the optimal route is selected using the recursive route searching algorithm. The LSTM predictor can reflect the impact of short-term demand and supply on the pickup probabilities. Meanwhile, the optimal virtual station v_i^* and its fastest driving route connecting r_i and v_i^* are found in Steps 6 and 7. Steps 8 and 9 provide the optimal searching route by comparing the expected profit for the route searching strategy G_i^{s*} and for the virtual station waiting strategy G_i^{w*}. After we provide the optimal driving route for taxi i, the taxi supply along the route is added by one and we go back to Step 1 to serve the $(i+1)$th taxi in the queue.

By considering the previously recommended vacant taxis, the dynamic pickup probability predictor can automatically capture the potential taxi oversupply by updating the pickup probability. Therefore, the searching routes of the queued taxis can vary from those of the scheduled vacant taxis to avoid recommendation overload.

Algorithm 2 (SRRS ($\mathcal{C}$))
Input: $\mathcal{C}$: a sequence of vacant taxis to be served;
Output: $\{R_i^* | \ \forall i \in \mathcal{C}\}$: optimal searching routes;

1: **Pop** Taxi $i \in \mathcal{C}$ at query time t_i and query segment r_i;
2: Construct recursion tree Υ_{r_i} according to the current searching-start segment r_i and searching time limit τ;
3: Update road network parameters $(v_{r,t_i}, o_{r,t_i}, t_r)$ for all $r \in \mathcal{G}$;
4: **Predict** pickup probability P_r for all $r \in \Upsilon_{r_i}$;
5: **Find** the optimal searching route: $R_i^{s*}, G_i^{s*} =$**RRS**(r_i, Υ_{r_i});
6: **Find** the optimal virtual station and the fastest driving route:
7: $v_i^*, R_i^{w*} = \arg\max_{v \in \mathcal{V}, R \in \mathcal{S}_{r_i}} \frac{1}{T_R + T_v} e_v$;
8: **Recommend** the optimal searching route from searching or waiting strategy:
9: $R_i^* = \arg\max_{R_i^* \in \{R_i^{s*}, R_i^{w*}\}} \left\{ \frac{1}{\sum_{r \in R_i^{s*}} t_r} G_i^{s*}, \frac{1}{T_{R_i^{w*}} + T_{v_i^*}} e_{v_i^*} \right\}$
10: **Update** taxi supply along route R_i^*;
11: Go to Step 1 to recommend the searching route for the next vacant taxi;

6. Numerical Experiments and Discussions

In this section, we demonstrate the efficiency and effectiveness of the proposed sequential route recommender system with extensive experiments using the Beijing Taxi data. All experiments were conducted on a server with 2X 10-core Intel Xeon Gold 5215 Processor, 1 TB RAM, and 10X 2080Ti GPUs. The Python scripts for all numerical experiments presented in this section can be found at the GitHub site of this paper (Liu et al. 2023).

6.1. Road Segment Pickup Probability Prediction

6.1.1. Baselines and Evaluation Metrics. The optimal hyperparameters of the proposed GCN-LSTM model are found through random searching (Bergstra and Bengio 2012). Specifically, we use one layer of graph convolution with dimension 32 and one layer of LSTM with dimension 16. The learning rate is 0.01, and the batch size is 8. To validate the effectiveness of the pickup probability predictor, the following baseline methods are selected for performance comparison:

1. **Classic Regressors.** The random forest (RF) (Breiman 2001) and gradient boosting trees (GBT) (Friedman 2002), two ensemble models for machine learning tasks, are used as classic baselines. They have been proven to be robust and effective but cannot deal with temporal and topological data by design (Maclin and Opitz 1997).

2. **Autoregressive Models.** The ARIMA and ARIMA with exogenous (ARIMA-X), two autoregressive models (Box et al. 2015), are used as linear autoregressive baselines. These two methods mainly focus on discovering temporal patterns from historical data.

3. **Nonlinear Autoregressors**. The nonlinear autoregressive network with exogenous inputs (NARX) and LSTM (Hochreiter and Schmidhuber 1997) have the ability to process the sequential data by modeling a nonlinear relationship between the time series values and exogenous variables.

4. **GNN-based Methods.** The MTGNN (Wu et al. 2020b) and GaAN (Zhang et al. 2018), two methods designed for traffic speed prediction, have the ability to learn spatiotemporal dynamics of road networks.

5. **Benchmark Models.** The historical average method from Qu et al. (2014) and the deep LSTM from Yu et al. (2017) are two latest benchmark models for the taxi recommendation in the literature.

The prediction accuracy is measured on an hourly basis using the mean absolute error (MAE) metric, which measures the absolute error of pickup probability prediction. Furthermore, we calculate the weighted average of the *MAE* (*wMAE*), weighted by the total number of trips during each time slot (denoted as $W(t)$):

$$MAE(t) = \frac{1}{N}\frac{1}{E}\sum_{i=1}^{N}\sum_{e=1}^{E} |\hat{p}_i^e(t) - p_i^e(t)|, \quad wMAE(t) = \frac{\sum_{t=1}^{24} W(t)MAE(t)}{\sum_{t=1}^{24} W(t)},$$

where $\hat{p}_i^e(t)$ represents the predicted pickup probability of segment i during tth time slot on the eth testing day. From the seven-day data set, we used the first four days as the training set and the remaining three days as the testing set. The evaluation follows the rolling forecasting for next-time-slot prediction. That is, after we predict the next-time-slot pickup probability, the data are included in our training set and the model is updated.

6.1.2. Experimental Results. We assess the performance of the proposed GCN-LSTM model by comparing it with the baselines using the same training and testing data set. Table 3 summarizes the performance comparison in

Table 3. Performance Comparison of Different Pickup Probability Predictors

		Weighted average MAE (wMAE)				Computational time (s)	
Class	Methodology	24 hours	Morning	Afternoon	Evening	Training	Execution
Benchmark	Historical mean	0.2237	0.1520	0.1478	0.1692	38a	10[a]
	Deep LSTM	0.1233	0.1069	0.1001	0.1138	1,104[b]	115[b]
Ensemble Methods	Random forest	0.2208	0.1921	0.1600	0.2179	218[a]	18[a]
	Gradient boosting	0.2497	0.1581	0.1413	0.1697	469[a]	33[a]
Autoregression	ARIMA	0.1721	0.1488	0.1308	0.1894	1,328[a]	105[a]
	ARIMA-X	0.1216	0.1075	0.1003	0.1412	1,609[a]	161[a]
Nonlinear Autogression	NARX	0.1640	0.1411	0.1398	0.1653	841[b]	46[b]
	LSTM	0.1588	0.1319	0.1321	0.1601	1,006[b]	84[b]
GNN-based Methods	MTGNN	0.1184	0.1206	0.1421	0.1278	1,022[b]	32[b]
	GaAN	0.1205	0.1028	0.1017	0.1112	327[b]	26[b]
Proposed method	GCN-LSTM	0.1112	0.0913	0.0902	0.1039	305[b]	13[b]

[a]Model trained and computed in CPU environment.
[b]Model trained and computed with GPU acceleration.

terms of *wMAE* defined previously. To test the robustness of the methods, we design four testing categories by computing *wMAE* over different time periods, including full day (24 hours), morning peak hours, afternoon non-peak hours, and evening peak hours. As seen, the proposed GCN-LSTM model achieves the lowest *wMAE* compared with all baselines across four testing categories. It shows the superiority of the proposed GCN-LSTM model compared with baselines and the robustness of its performance. The performance comparison between different methods also offer justifications on why the proposed GCN-LSTM model is superior by design.

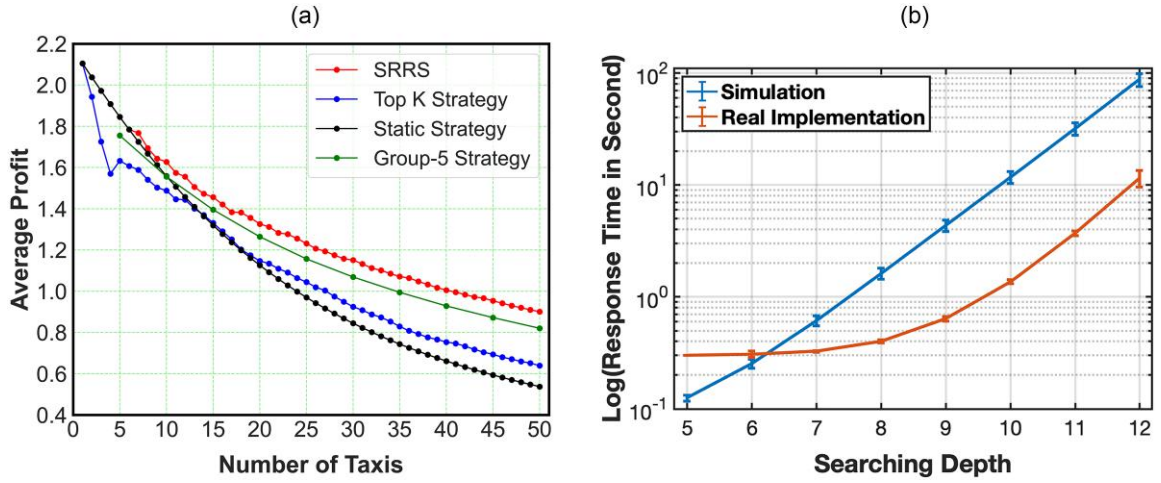
- The autoregression methods outperform the ensemble methods, which indicates the importance of time dependencies in predicting pickup probabilities and justifies the inclusion of LSTM in the proposed model.
- The proposed GCN-LSTM model outperforms LSTM, ARIMA-X, and deep LSTM methods, indicating that GCN can better capture the taxi activities using a network structure. The importance of graph modeling in predicting pickup probabilities is further verified by the good performance of other GNN-based models, such as MTGNN and GaAN.
- The proposed GCN-LSTM model outperforms MTGNN in all four testing categories. This shows the main difference in model design that MTGNN tries to learn the graph adjacency matrix based on time series data, requiring extremely long time series (around 10,000) to achieve its optimal performance. In our data set, the length of time series is at hundreds level, which does not support a good training of the MTGNN model. Meanwhile, the shorter time series also impact the graph convolution performance of MTGNN because it does not explicitly model the short-term and long-term impact as our framework.
- Compared with the GaAN model, the proposed GCN-LSTM model benefits from the choice of the LSTM unit, which provides better temporal dynamics modeling of a sequence of graph convolution networks.

Finally, we compare the computational efficiency of the proposed GCN-LSTM model with benchmark models, measured by the training time and execution time of the models. Specifically, the training time includes the computational time for formatting training data and model training until convergence, and the execution time includes the computational time for formatting input data and making predictions. As some baseline algorithms do not support GPU acceleration, the traditional machine learning methods and autoregressive methods were computed in the CPU environment and the deep learning methods were computed using GPU. The computational efficiency comparison is reported in the last column of Table 3. It is seen that the proposed GCN-LSTM model can be trained in 305 seconds, indicating that the offline road network analysis can be conducted with the newly observed data at a high frequency of every five minutes. A well-trained GCN-LSTM model can update the distribution of pickup probability in the searching area in 13 sets, which is applicable for the online route recommendation. It takes a much longer time for the MTGNN model to converge, whereas the GaAN and GCN-LSTM models are more efficient using the traffic network data. Compared with the NARX and LSTM methods that treat road segments independently, the GCN-LSTM method is more efficient by batch learning the model parameters in the network.

6.2. Effectiveness on Mitigating Recommendation Overload

During rush hours, a number of taxis may dropoff on the same road segment and then submit queries for the next searching route during the same time period. For example, in the evening rush hours, vacant taxis may start searching from the residential areas after completing the commuting trips. As a result, we can observe a severe taxi oversupply in some regions. Here, we provide a case study to show the effectiveness of our sequential route searching

Figure 8. (Color online) Performance Comparison of Different Routing Strategies for a Balanced Recommendation



Notes. (a) Performance comparison. (b) Query response speed test.

strategy in addressing this issue. In this case study, starting from 5:30 p.m., 50 taxis started searching for the next passenger from the same road segment (indexed 63819) within a five-minute period. The searching depth is fixed to be eight. After the system provides a route recommendation for the ith taxi, the taxi supply and the corresponding pickup probability along this recommended route are immediately updated using the trained GCN-LSTM predictor to provide a balanced sequential route recommendation.

We compare the proposed sequential route recommendation with the top K strategy (Qu et al. 2014), the static strategy without considering recommendation overload, and a simple group recommendation. The profit per minute of searching time is used to evaluate the model performance. The top K strategy provides a different route by comparing the route similarities with the scheduled routes, while the static strategy does not consider recommendation overload and the same routes will be recommended. The group-five strategy provides an intuitive batch recommendation for every group of five queries. The top five most profitable routes will be recommended to a group of five taxis. The expected profit is estimated after a batch recommendation is made, and then the distribution of pickup probability is updated.

Figure 8(a) shows the average profit for K scheduled taxis; that is, after we provide route recommendations for K taxis, their average profit is used as the evaluation metric. As seen in Figure 8(a), the average profit decreases as more taxis start searching in the same segment. The sequential route recommendation can provide a balanced recommendation for a queue of 50 taxis with an average profit of 0.9003 CNY/minute. The static strategy has a serious recommendation overload issue with an average profit of 0.537 CNY/minute for 50 taxis. The top K strategy selects the recommended routes according to route similarity while neglecting potential profit. The expected profit decreases significantly at the beginning in order to avoid repeating routes, but overall it provides a better solution than the static strategy, with an average profit of 0.6388 CNY/minute for 50 taxis. The group-five strategy with an update frequency of every five queries achieves the second-best performance with an average profit of 0.8202 CNY/minute for 50 taxis, which provides an alternative solution to the sequential search strategy that requires an update frequency of once per query.

Furthermore, Figure 9 presents the top six recommended routes using the proposed method. Specifically, the route in Figure 9(a) is recommended for the first six taxis and the 16th and 26th taxis. After six taxis adopt the first route, it is no longer optimal for the subsequent taxis waiting in the queue. As a result, for taxis 7–9, the route in Figure 9(b) is recommended. Overall, the sequential route recommendation with dynamic pickup probability update can help address the recommendation overload issue when there is an oversupply.

6.3. Evaluation on Sequential Recommender Systems

6.3.1. Experimental Setting. To validate the effectiveness of the proposed mobile route recommender system, a total of 108,793 real-world searching route queries during the fifth to seventh days (out of the seven-day data set) within the third ring road circle in Beijing were used as the testing data. Each query record includes the actual searching-start time t_0, searching-start location r_0, actual searching route $R^*_{r_0}$, searching time Δ, and earning of this pickup order e. Figure 10(a) presents the searching-start point distribution.

Figure 9. (Color online) Route Recommendation for a Queue of Taxis



Notes. (a) Recommended route for taxis 1–6, 16, and 26. (b) Recommended route for taxis 7–9, 11, and 28. (c) Recommended route for taxis 10 and 22. (d) Recommended route for taxis 12–15 and 29. (e) Recommended route for taxis 17 and 25. (f) Recommended route for taxi 18.

It is not trivial to conduct performance comparisons using the actual data, because we only know what route a taxi driver actually took but not the routes that the taxi could have taken. To establish fair comparisons, we control the taxi demand-supply distribution by counting the number of pickup orders o_r and the number of vacant taxis v_r for all road segments in its searching road network $r \in \mathcal{G}_{r_0}$. Vacant taxis at time t_0 with unfinished searching routes in the next 10 minutes were considered as previously scheduled taxis. The taxi demand-supply distribution is then fed into the GCN-LSTM model to update the pickup probability distribution in this searching road network $\mathcal{G}_{r_0}$. To control the searching time, if the actual searching time Δ is less than τ (10 minutes in our experiments), we use Δ as the searching time to make a recommendation. Otherwise, we consider the taxi out-of-service and filter the record.

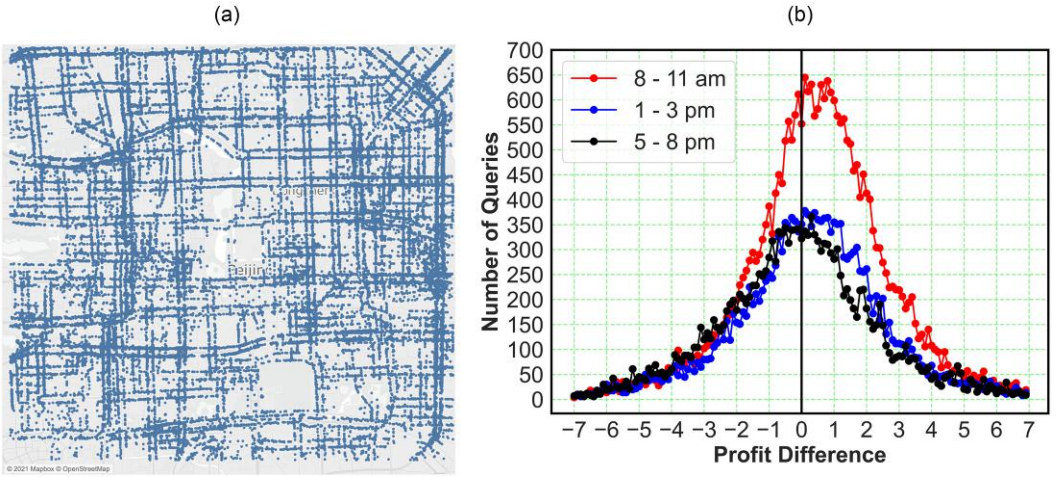
For a single query, three metrics are computed for performance comparison:

1. **Actual Unit Profit (AUP)**. Each actual taxi query record has the actual earning by taking this actual route. We compute the actual profit per minute made by this actual taxi on this trip as a baseline: $G_A^* = \frac{e}{\Delta}$.

2. **Expected Unit Profit (EUP) of Actual Routes**. For each actual route that a taxi took, we compute the expected profit estimated using the predicted pickup probability, as well as the attendant expected profit per minute. Actual routes in the historical data are routes that end up with a successful pickup. Thus, the expected unit profit of actual routes should be smaller than the actual unit profit, since the former metric takes into account the probability that the taxi does not successfully pickup a passenger after finishing the route. Hence, this metric is a fair baseline for our method.

3. **Expected Unit Profit (EUP) of Recommended Routes**. Taking the same starting scenario of an actual route, we run the proposed method and compute the expected profit if the taxi takes the route recommended by our

Figure 10. (Color online) Experiment of Sequential Route Recommendation for Vacant Taxis



Notes. (a) Vacant taxi query distribution. (b) Profit difference between SRRS and the benchmark model.

method, as well as the attendant expected unit profit. This metric represents the expected unit profit when the taxis take the routes recommended by the proposed method.

In addition, we test the proposed algorithm when using the minimum searching time (MST) and minimum searching distance (MSD) (Xiao et al. 2020) objectives, by replacing the earning e_r on road segment r with driving time T_r and distance L_r, respectively. Their objective functions are formulated as follows:

$$G_{MST}(R; r_0, t_0) = p_{r_0} T_{r_0} + \sum_{i=1}^{n} \prod_{j=0}^{i-1} (1 - p_{r_j}) p_{r_i} T_{r_i}, \tag{13}$$

$$G_{MSD}(R; r_0, t_0) = p_{r_0} L_{r_0} + \sum_{i=1}^{n} \prod_{j=0}^{i-1} (1 - p_{r_j}) p_{r_i} L_{r_i}. \tag{14}$$

The expected profit of the optimal routes with minimum searching time and minimum searching distance are then calculated as two baselines.

Finally, we test the proposed method and the two alternative methods using MST and MSD objectives under both static and dynamic modes. The dynamic mode means we dynamically update the pickup probability predictions when making sequential recommendations, while the static mode means we use the fixed historical average pickup probability without dynamic updates and make sequential recommendation using the strategy proposed by Qu et al. (2014).

6.3.2. Experimental Results. Figure 10(b) presents the histogram of performance comparison between our proposed cost-effective sequential route recommender system (C-SRR) and the AUP, where the x axis is the difference between the profits of the recommended results and the taxi drivers' actual profits. It is seen that the profit of our recommended routes outperforms the profit of the actual routes chosen by the taxi drivers.

We then compute the three performance metrics under different benchmarks introduced previously. Table 4 summarizes the performance comparisons. It is seen that the expected unit profit of taking actual routes is 2.896 CNY/minute, which is lower than the actual unit profit 3.174 CNY/minute, because the latter only computes routes ending up with a successful pickup. In comparison, the expected unit profit of taking routes recommended by the proposed C-SRR strategy with dynamic pickup probability update can achieve an expected unit profit of 3.560 CNY/minute, a 22.93% of improvement from the expected unit profit of actual routes. It shows that taking the recommended routes are expected to increase the unit profit of taxi drivers. Although we control the searching time limit of $\tau = 10$ minutes, taxi drivers can set a smaller searching time limit if they only care about the short-term expected earnings or want to restart the search earlier if no passenger is found. By setting the searching time limit to $\tau = 8$ minutes, the EUP of actual routes within 8 minutes is 3.374 CNY/minute. Taking these real searching events with their actual searching time and searching-start road segments, the C-SRR strategy can provide a EUP of recommended routes of 3.863 CNY/minute, a 14.49% improvement from the EUP of actual routes.

Table 4. Profit Comparison of Different Recommendation Strategies in CNY per Minute

Prediction model	AUP of actual routes	EUP of actual routes	EUP of recommended routes (MST)	EUP of recommended routes (MSD)	EUP of recommended routes (C-SRR)
Static model	3.174	2.896	2.903	2.518	3.107
Dynamic model			3.335	3.221	**3.560**

Notes. The bold number indicates the largest number in this table for performance comparison. It is obtained by the C-SRR strategy in a dynamic model.

Furthermore, the performance comparison between the static and dynamic models indicates that we can achieve a higher expected profit by considering the real-time recommendation overload (dynamic updates of pickup probability prediction). It is worth noticing that static strategies can have worse performance than taking the actual taxi routes, indicating experienced taxi drivers may have the knowledge to avoid oversupplied regions. Even using the objectives of MST and MSD, adopting the dynamic pickup probability predictor could have increased the expected unit profit by 14.88% and 27.92%, respectively. The minimum searching distance strategy (using the MSD objective) outperforms by the minimum searching time strategy (using the MSD objective), because the latter ignores traffic congestion that may cause a long searching time and a low earning per minute. Both strategies are worse than the proposed method using the objective in Equation (1), indicating that the C-SRR is more profitable for taxi drivers by selecting road segments with higher earnings per trip. It is also important to note that, the minimum searching time strategy can be more profitable for on-demand drivers such as Uber or DiDi drivers who take cash bonuses by finishing more orders.

6.3.3. Efficiency Test. Finally, to evaluate the computing efficiency of the proposed C-SRR system, we use the case described in Section 6.2 when there is a taxi oversupply and the case of a three-day system implementation for real-world queries. Because the depth of the recursion tree affects the computing efficiency the most, we measure the query response time by varying the tree depth from 5 to 12 (roughly equivalent to 5 to 18 minutes of searching time limit τ). The result of the query response times is presented in Figure 8(b). The bottom line shows the query response time for a queue of taxis starting searching on the same road network, with the bar on each data point representing the standard deviation. The computational time for the recursion tree construction is excluded and the response time indicates the efficiency of route optimization. It can be seen that the recursive searching algorithm is efficient and the response time is less than 1 second if the searching depth is no more than nine segments. However, the computational time increases exponentially with search depth and the response time can go up to 10 seconds when the search depth reaches 12. The upper line in Figure 8(b) presents the response time of a three-day implementation. The response time increases as the search depth increases, but the query response for a search depth of 8 is less than 1.1 seconds, which is acceptable for a real-time online recommender system.

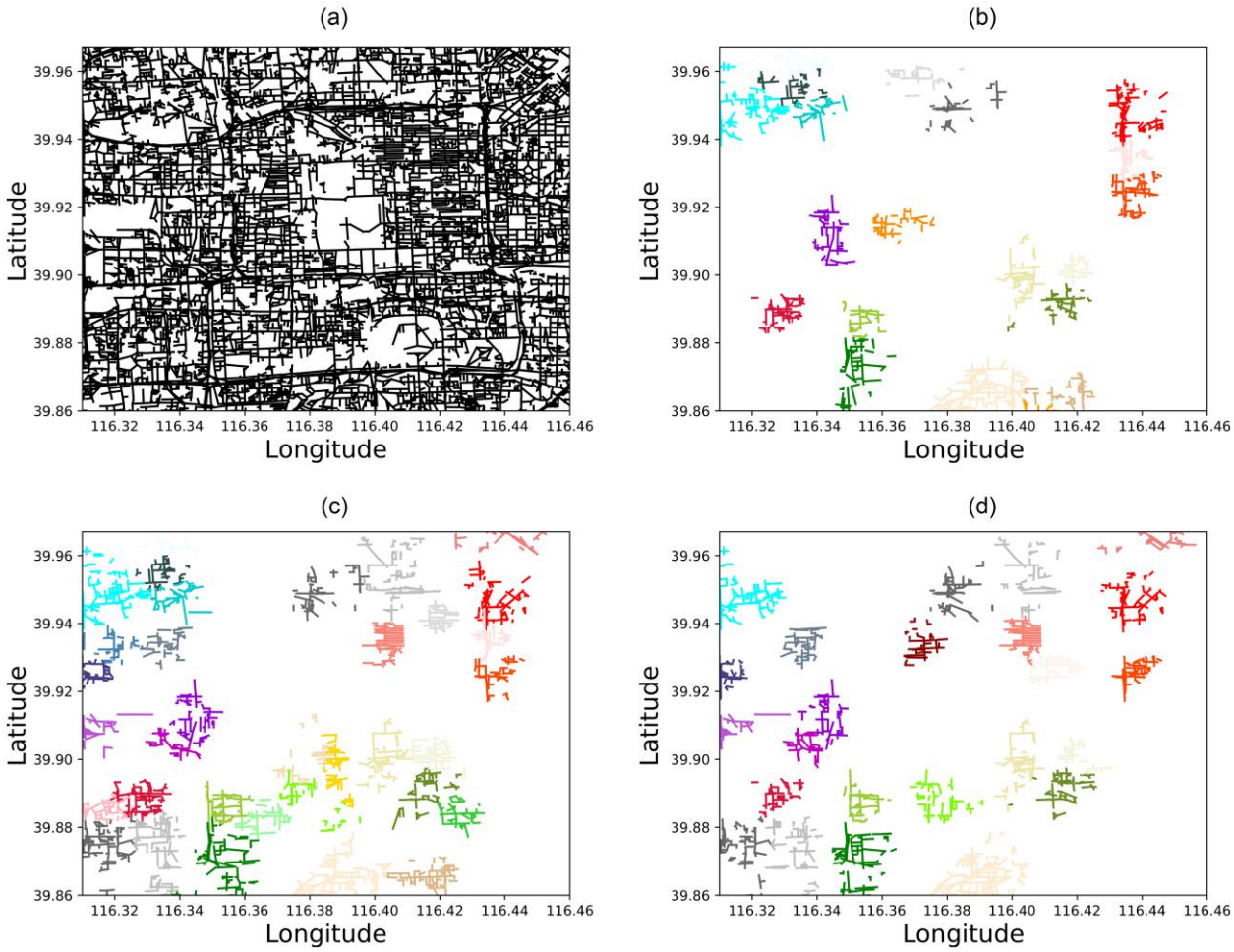
6.4. Active Region of Virtual Stations

Sometimes, waiting in a virtual station can be more profitable than searching along a route, especially when the current searching-start segment of a taxi is near such a virtual station. On the one hand, if the current location is far away from any virtual stations, the taxi will waste gas and time driving to a virtual station, resulting in a low net profit per minute. In those cases, the waiting strategy is inferior to the route searching strategy. On the other hand, during certain time periods and when the searching-start segments are within certain regions, the waiting strategy could be superior to the route searching strategy. The set of searching-start segments $\mathcal{R}_w$ and time periods $\mathcal{T}_w$ preferred for adopting the waiting strategy are discovered as so-called active regions and active time zone of virtual stations. We extract the recommendation results for the seventh day to find out whether a searching-start query is responded to with a virtual station instead of a searching route in the next 10 minutes. If there is at least one query recommended with a virtual station, we label this searching-start road segment as an active waiting segment. The searching-start segments recommended with the same virtual station form an active waiting region. Figure 11 presents the active waiting region of virtual stations during three different time periods, with the color representing the regions of different virtual stations. As seen, the virtual stations and active regions change during the day. Further, most of the active virtual stations are actual taxi stands or are located near hospitals, museums, hotels, schools, and sports centers in Beijing.

7. Concluding Remarks

This paper proposes a cost-effective recommender system for taxi drivers to maximize their profits by providing profitable searching routes that avoid recommendation overload. Specifically, we integrate a recursive route searching

Figure 11. (Color online) Active Regions of Virtual Stations During Different Time Periods



Notes. (a) Base map. (b) Active regions, 8–11 a.m. (c) Active regions, 1–3 p.m. (d) Active regions, 5–8 p.m.

strategy with a segment-level pickup probability predictor based on the GCN-LSTM model. GCN is designed to capture taxi demand-supply distribution as a topological structure in a road network, and the LSTM model is designed to model the temporal dynamics of the road network. Based on a timely and precise prediction of pickup probabilities, the recommender system can achieve the balance of demand-and-supply mobility and avoid recommendation overload. To improve the computational efficiency of route optimization, we further propose a recursive route searching algorithm to support the real-time recommender system for a sequence of taxi drivers. The algorithm takes the prediction result as an input to estimate and compare the potential profits of candidate routes, significantly reducing the size of route searching space. In addition, the algorithm parameters are updated dynamically according to the previously scheduled driving routes to avoid recommendation overload. The route searching strategy is then compared with the waiting strategy by discovering profitable virtual stations, incentivized by experienced taxi drivers who prefer waiting for passengers at hot spots. Our data have shown that such virtual stations change over time during the day. Finally, we conduct extensive experiments on real-world data sets collected from Beijing and the experimental results clearly validate the effectiveness of the proposed recommender system.

This paper can be extended in several future directions. The proposed recommender system considers the expected earnings of the next pickup for a single taxi driver searching in a localized region. The localized sequential search follows the first-come-first-serve policy that responds to route queries reactively. It would be beneficial to design a multistep long-term taxi demand predictor to proactively guide taxis to the regions with high taxi demand if the nearby road network has a very low pickup probability during off-peak hours. In addition, the batch recommendation can be an alternative strategy to the sequential search during peak hours. The batch recommendation seeks a global earning for a group of taxis but requires drivers to wait for a batch of route queries from this group. When the group of queries emerges in a very short period of time in the same region during peak hours, the batch

recommendation may provide a good solution without a high query response latency. A globally optimal solution may be possible if the pickup probability predictor can be integrated into an optimization model for group route assignment. Furthermore, in our study, the influence of a recommendation on future demand distribution can be fully captured in the consequent queries. It would be desirable to address the potential concurrency issue that many taxi drivers are sharing and updating information in the same search network within a short period of time. The concurrency control problem can be exacerbated in a distributed database system where the traffic data are usually stored. As a future study, we can impose concurrency control solutions in a distributed database system to further improve the computational efficiency and effectiveness of the route recommender systems.

The proposed sequential route recommendation methods can also be generalized to facilitate the development of other mobile recommender systems. Although the proposed system is developed to provide street-hail services, the combination of a GCN-LSTM predictor and recursive route recommendation algorithm can help improve the work efficiency of the on-demand ride-hailing systems (Alonso-Mora et al. 2017, Braverman et al. 2019) by recommending vacant sharing cars to drive to the road segments with a high probability of sharing orders and high earnings. Because the external information, such as prices, may affect the demand-supply distribution in the on-demand system, the proposed GCN-LSTM predictor can be further extended to help solve the pricing optimization problems (Bimpikis et al. 2019, Yang et al. 2020). The sequential route recommendation system can also benefit crowdsourced delivery services (Alnaggar et al. 2021) by recommending driving and delivering routes for new orders while avoiding the competition of delivery agents in the same region.

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